

An IoT-Based Smart Helmet for Multimodal Rider Health Monitoring and Proactive Risk Detection.

Abstract—Motorcyclists and cyclists encounter elevated accident risks stemming from environmental hazards, physiological strain, cognitive stress, and dynamic road conditions. Conventional helmets offer only passive physical protection, lacking real-time monitoring and predictive capabilities. This paper presents an integrated IoT-enabled smart helmet system that delivers multimodal rider monitoring, real-time risk detection, and proactive safety interventions. The system fuses data from multiple sensor streams including EEG (NeuroSky headset), heart rate and body temperature (ESP32-based smartwatch), IMU (accelerometer and gyroscope), and GPS processed through a hybrid edge-mobile architecture. Four tightly coupled modules form the core of the system: (1) real-time health monitoring with AI-driven heart attack risk prediction using a TensorFlow Lite model; (2) EEG-based cognitive stress and fatigue detection combining machine learning inference on raw brainwave signals with physiological fallback mechanisms; (3) live IMU/GPS-driven risky event detection (sharp turns, harsh braking, aggressive leans) with post-journey risk assessment and danger zone clustering; and (4) route-level danger zone prediction that integrates historical ride data, weather conditions, traffic information, and a dedicated TFLite risk model. The system provides context-aware multilingual voice guidance, smartwatch notifications, hands-free emergency support, and personalized post-ride safety reports incorporating spatial risk visualization and performance analytics. Extensive real-world testing demonstrates high accuracy in stress classification, physiological risk assessment, and danger zone identification, significantly enhancing early warning capabilities compared to traditional approaches. By shifting from passive protection to intelligent, predictive rider safety, the proposed framework contributes a practical and scalable solution for reducing motorcycle accidents through data-driven decision support in real-world riding conditions.

Keywords—Smart helmet, Internet of Things (IoT), multimodal sensor fusion, electroencephalography (EEG), stress and fatigue detection, machine learning, danger zone prediction, edge computing, real-time risk assessment, post-journey analysis.

I. INTRODUCTION

Road traffic crashes remain a critical global public health issue. The World Health Organization (WHO) estimates that approximately 1.19 million people die each year due to road traffic injuries, with more than half of these fatalities involving vulnerable road users such as pedestrians, cyclists, and motorcyclists, particularly in low- and middle-income countries [1]. Motorcyclists face heightened risks due to their direct exposure to environmental hazards, variable traffic conditions, and the interplay between physiological stressors and cognitive fatigue, all of which can impair judgment and reaction times. Traditional helmets offer essential passive protection through impact absorption but lack the capability to monitor the rider's physiological condition, detect cognitive impairment, or provide proactive safety interventions. Recent advances in wearable sensors, Internet of Things (IoT) technology, edge computing, and machine learning have enabled the development of intelligent helmets

that go beyond passive safety [2], [4]. Existing systems have demonstrated the feasibility of post-crash emergency alerting using GPS and GSM modules [7], basic physiological monitoring [8], [10], fatigue detection via EEG [3], and rudimentary accident-avoidance mechanisms [12], [13]. However, most prior solutions address only isolated aspects such as vital signs monitoring or simple crash notification while falling short of delivering a unified, multimodal platform that simultaneously handles real-time cardiac risk, cognitive stress, motion-based risk events, and geospatial danger zone prediction. This paper presents a comprehensive IoT-enabled smart helmet system that integrates multimodal sensing and intelligent analytics to provide proactive rider safety. The system fuses data from EEG, heart rate, body temperature, inertial measurement unit (IMU), and GPS sensors through an ESP32-based edge platform, with advanced processing performed on both edge devices and a companion mobile application. By combining real-time monitoring, machine learning-driven risk assessment, and post-ride analytics, the proposed system shifts the paradigm from reactive protection to predictive, personalized safety management. The main contributions of this work include the development of a multimodal sensing architecture that integrates physiological (heart rate, temperature, EEG), motion (IMU), and geospatial (GPS) data streams for holistic rider state monitoring. Edge-optimized TensorFlow Lite models are developed for real-time cardiac risk assessment and EEG-based cognitive stress and fatigue detection. A hybrid danger zone detection framework is proposed that combines live IMU/GPS event detection, historical ride data, environmental context (weather and traffic), and spatial clustering techniques to identify and alert riders to high-risk road segments. Finally, a cross-platform Flutter mobile application is implemented featuring real-time visualization, multilingual voice guidance (English and Sinhala), smartwatch integration, and comprehensive post-journey risk reports with spatial analytics. Through these contributions, the proposed system delivers timely, context-aware interventions while generating actionable longitudinal insights for riders. The remainder of this paper is organized as follows: Section II describes the overall system architecture, Section III details the implementation of the core monitoring components, Section IV presents the mobile application and user interface, Section V evaluates the performance of the prototype, and Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

Research on smart helmets has progressed from basic post-crash telemetry systems to more sophisticated intelligent platforms. Early efforts primarily focused on emergency notification using GSM modules and GPS tracking following accidents [7], [14], [15]. Subsequent studies incorporated physiological sensors for health monitoring, particularly

heart rate and body temperature, enabling basic real-time rider state assessment [8], [10], [11]. More recent works have explored the integration of artificial intelligence for accident prevention and risk prediction. Kim *et al.* [12] proposed an AI-integrated smart helmet for motorcycle safety, while Khan *et al.* [13] developed an IoT-based system with a dedicated motorbike unit to enhance rider connectivity and safety. Parallel research in wearable electroencephalography (EEG) has demonstrated the potential of brainwave analysis for fatigue and cognitive state monitoring in dynamic environments [2], [3]. Sensor fusion techniques combining multiple modalities have also shown improved performance over single-sensor approaches [13]. Despite these advancements, significant gaps remain. Most existing systems address only isolated aspects such as post-crash alerting, basic vital signs monitoring, or simple environmental sensing rather than providing a unified, real-time, multimodal framework. Few solutions integrate cognitive stress detection via EEG, proactive geospatial danger zone prediction, and comprehensive post-ride analytics. Moreover, limited attention has been given to low-cost, edge-optimized implementations suitable for low- and middle-income countries or to multilingual voice guidance tailored to regional contexts. The present work addresses these limitations by developing a comprehensive smart helmet system that simultaneously monitors physiological health, cognitive stress, motion dynamics, and route-based risks within a cohesive edge-mobile architecture, supported by personalized post-ride analytics and multilingual alerts.

III. SYSTEM ARCHITECTURE

The proposed smart helmet system is a multimodal IoT-enabled wearable platform designed to provide continuous monitoring, real-time risk assessment, and proactive safety interventions for motorcyclists and cyclists. The architecture integrates hardware sensors, embedded edge processing, wireless communication, and a companion mobile application. It is organized into four primary layers: sensing, edge processing, communication, and cloud-mobile analytics. The hardware layer consists of a comprehensive suite of sensors coordinated by an ESP32 microcontroller. Physiological sensing is achieved through the MAX30102 optical sensor for heart rate, SpO₂, and heart rate variability (HRV) monitoring, while the MLX90614 infrared thermometer measures body and ambient temperature for heat stress detection. Cognitive monitoring is performed using a single-channel EEG sensor embedded in the helmet lining. Motion and riding behavior are captured by the MPU6050 6-axis IMU operating at 100 Hz for detecting sharp turns, harsh braking, aggressive leans, and falls, supported by the NEO-6M GPS module for real-time positioning and route tracking with dead reckoning capability. The ESP32 serves as the central edge processor, responsible for sensor data acquisition, real-time filtering, anomaly detection, and Bluetooth Low Energy (BLE) transmission. On-device intelligence is supported by TensorFlow Lite models, while voice interaction is enabled through PicoVoice and Google TTS for multilingual alerts. The software architecture is divided between embedded firmware on the ESP32 and the cross-platform Flutter mobile application. The ESP32 firmware, developed using ESP-IDF

and Arduino IDE, handles low-level sensor acquisition, Butterworth filtering, and preliminary threshold-based event detection. The Flutter-based mobile application acts as the central intelligence layer, performing advanced machine learning inference using TensorFlow Lite models for cardiac risk assessment, EEG-based stress and fatigue detection, and IMU-driven risky event classification. It further integrates external APIs for weather and traffic data to support route-level danger zone prediction, executes spatial risk analysis, manages proximity alerts, and synchronizes telemetry with Firebase Firestore. The system operates across three operational phases. In the pre-ride phase, the mobile application analyzes historical ride data and environmental conditions (weather and traffic) to generate route risk assessments and safety recommendations. During the ride phase, raw sensor data streams to the ESP32 for initial processing and is then transmitted via BLE to the mobile application, where multimodal fusion, real-time inference, voice alerts, and danger zone proximity warnings are executed. In the post-ride phase, all telemetry is synchronized to Firebase Firestore, enriched with contextual data, and processed to compute safety scores, cluster danger zones, and generate comprehensive personalized risk reports.

IV. COMPONENT IMPLEMENTATION

The smart helmet system comprises four tightly integrated modules that work collaboratively to deliver comprehensive rider safety. The Health Monitoring Module serves as the real-time physiological tracking and emergency response system. It utilizes a dual-core ESP32 microcontroller interfaced with the MAX30102 photoplethysmography sensor for heart rate and SpO₂ monitoring and the MLX90614 infrared sensor for body temperature measurement. A five-point moving average filter mitigates motion artifacts before data is packaged into JSON payloads and transmitted via BLE every 500 ms to the Flutter mobile application. The application runs an on-device TensorFlow Lite model that processes a seven-feature vector (heart rate, temperature, age, gender, weight, height, and BMI) to predict cardiac risk. This risk score is dynamically adjusted using an EEG-based emotion multiplier. When the adjusted risk exceeds 70% for 30 seconds, the system automatically triggers an SOS emergency protocol, sending SMS and email notifications with vital signs, risk level, and Google Maps location to emergency contacts, while logging all data to Firebase Firestore for longitudinal analytics and PDF report generation. The Stress Detection Module monitors the rider's cognitive state in real-time using a hybrid EEG and physiological approach. It interfaces with the NeuroSky MindWave Mobile 2 EEG headset (HR-S0C1913) via the ThinkGear protocol over Bluetooth SPP, capturing raw EEG at 512 Hz and attention/meditation metrics at 1 Hz. Raw signals undergo bandpass filtering, Fast Fourier Transform (FFT), and spectral power analysis across Delta, Theta, Alpha, Beta, and Gamma bands. A pre-trained TensorFlow Lite model performs inference on a 14-dimensional feature vector, producing a stress probability score (0.0–1.0) with under 50 ms latency. When EEG signal quality degrades, the system automatically switches to a physiological fallback using heart rate deviation and body temperature. A

persistence mechanism requires sustained high stress (>65%) for 30 seconds before triggering alerts. The module also includes fatigue detection based on low attention, high Theta/Alpha ratio, and reduced Beta activity. All readings are stored in Firebase Firestore every 10 seconds, and the user interface provides live EEG waveform, spectral power bars, and color-coded stress indicators. The Post-Journey Analysis Module (Member3) functions as both a real-time riding behavior monitor and a post-ride risk assessment engine. It receives data from the MPU6050 6-axis IMU at 10 Hz through the ESP32 helmet via BLE. Raw signals are preprocessed with yaw smoothing, turn rate calculation, and lean angle computation using $\text{leanAngle} = \text{atan2}(\text{accelY}, \text{accelX}) \times (180/\pi)$. Rule-based algorithms detect sharp turns (>100°/s), risky turns (>150°/s), harsh braking (>0.5g), and critical leans (>35°). A TensorFlow Lite model processes a 12-feature vector to output a continuous risk probability. State management is handled by JourneyProvider and RideSessionProvider for cross-module synchronization using a shared rideId. Upon ride completion, the system computes aggregate metrics and generates a detailed report on the JourneyReportScreen, featuring an interactive Google Map with GPS trajectory, color-coded event markers, clustered danger zones, and correlated stress data. The module supports PDF export and maintains low resource usage (35 MB memory, 12–15% battery per hour). The Danger Zone Detection Module provides proactive route risk intelligence. It combines historical journey data from Firebase Firestore, real-time weather from OpenWeatherMap API, traffic information, and live GPS. Historical risky events (turns and braking) are visualized as risk circles (25–60 m radius) on Google Maps. For new routes, the system performs segment-by-segment analysis using a TensorFlow Lite model with 27 features (weather, traffic, road type, etc.), color-coding the route polyline according to risk level. During active rides, it monitors proximity to historical risk zones within a 100-meter radius and delivers one-time voice alerts via FlutterTTS. The module includes dynamic safety tips and graceful degradation for API failures. It demonstrates efficient performance with sub-2-second historical data loading and under 50 ms inference per segment.

V. MOBILE APPLICATION AND ANALYTICS

The Flutter-based companion mobile application serves as the central user interface for real-time monitoring, proactive risk assessment, and comprehensive post-ride analytics. Developed for both Android and iOS platforms, it integrates data from all four core modules to deliver a seamless rider experience. The live dashboard (Member1 and Member2) provides real-time visualization of vital signs, including heart rate, body temperature, and cardiac risk levels through color-coded gauges. It simultaneously displays EEG-derived cognitive metrics such as stress level, fatigue index, and emotional state. An integrated map view overlays live location with proximity alerts for nearby danger zones. Prior to each journey, the pre-ride safety module (Member4) performs route risk prediction by combining historical ride data, current weather conditions, traffic information, and geospatial analysis. The application presents a color-coded route heatmap (red for high risk, amber for medium, and

green for low risk), an overall safety score, and alternative route suggestions when the predicted risk exceeds acceptable thresholds. Following ride completion, the post-ride analytics engine (Member3) automatically generates a detailed safety report within seconds. This report incorporates IMU-based risky event detection (sharp turns, harsh braking, and aggressive leans), clustered danger zones, physiological and cognitive summaries from Members 1 and 2, and route risk insights from Member 4. The system also maintains longitudinal analytics, allowing riders to review weekly and monthly trends in safety scores, stress patterns, fatigue occurrences, and environmental risk correlations. Data can be exported as PDF or CSV reports for personal records or sharing with fleet managers and insurers. The application further delivers personalized coaching by leveraging the rider's historical risk profile to generate context-aware recommendations and multilingual voice guidance in English or Sinhala. Users can customize notification thresholds, measurement units, and preferred language settings to suit individual needs.

VI. EVALUATION AND RESULTS

The Smart Helmet system integrates four AI-enabled modules targeting distinct aspects of rider safety. Table I presents a comparative performance summary of these modules.

Table I Performance Summary of Smart Helmet AI Modules

Component	Model Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Health Monitoring	Rule-based + Lightweight Classifier	85.0	82.0	88.0	85.0
Stress Detection	1D Convolutional Neural Network	73.99	85.0	71.4	70.8
Post-Journey Risk Assessment	Random Forest + Distilled Keras NN	74.65	43.0	48.0	45.0
Danger Zone Prediction	Dense Neural Network	52.00	—	—	—

Table II Performance Comparison Against Baselines

Component	Baseline Approach	Baseline Score (%)	Proposed Score (%)	Improvement
Health Monitoring	Threshold-based rules	~78.0	~85.0	+7.0%
Stress Detection	Conventional EEG classifiers	~68.0	70.8 (F1)	+2.8%
Post-Journey Risk Assessment	Standard Random Forest	64.0 (F1)	74.65 (Acc.)	+10.65%
Danger Zone Prediction	Random guessing	50.0	52.00	+2.0%

The Health Monitoring module achieved strong performance with an overall accuracy of 85.0% using a lightweight classifier supported by rule-based logic. For cardiac risk prediction, an XGBoost model initially reached 99.81% accuracy on a large test set. A distilled lightweight neural network was subsequently developed for mobile deployment, maintaining 89.2% accuracy and 0.934 AUC-ROC with an average inference latency of 47 ms, satisfying real-time requirements. The system reliably triggers emergency SOS alerts when risk exceeds 70% for 30 seconds. The Stress Detection module was evaluated on the SAM40 EEG dataset. After preprocessing and augmentation, the 1D Convolutional Neural Network achieved a best validation accuracy of 73.99%. On the held-out test set, the model recorded 85.0% precision and 71.4% recall for the stress class, with an overall accuracy of 69.1% and weighted F1-score of 70.8. The hybrid approach combining model output with band ratios and physiological fallback demonstrated stable real-time performance with end-to-end latency under 100 ms.

Table III Classification Performance of the Stress Detection Model

Class	Precision	Recall	F1-Score	Support
Relax	0.421	0.623	0.502	273
Stress	0.850	0.714	0.776	819
Weighted Average	0.743	0.691	0.708	1,092

The Post-Journey Risk Assessment module was trained and evaluated using the TU Wien IMU dataset for motorcyclist behavior. After feature engineering and balancing, the optimized Random Forest model achieved 74.65% accuracy, while the distilled Keras neural network reached 75.47% accuracy. Lean angle magnitude, resultant gyroscope, and yaw rate were identified as the most influential features. The final TensorFlow Lite model (10.6 KB) delivers real-time inference below 30 ms, enabling effective detection of risky riding events with low computational overhead. The Danger Zone Prediction module was trained on a synthetic dataset of 1,999 routes. Due to limitations in data quality, the Dense Neural Network achieved only 52.00% test accuracy, showing severe overfitting and no meaningful generalization. Although the machine learning component underperformed, the rule-based safety recommendation system based on weather, traffic, and road conditions remains functional and provides useful contextual guidance. This highlights the critical need for real-world labeled data for effective route risk prediction. Overall, the operational modules demonstrate strong potential for real-time deployment. Member2 and Member3 delivered competitive performance on real-world datasets, while the system as a whole maintains excellent efficiency with total model storage under 125 KB and sub-50 ms inference latency, validating the feasibility of the proposed multimodal smart helmet architecture.

VII. CONCLUSION

This paper presented a comprehensive IoT-enabled smart helmet system that transforms traditional passive protection into an intelligent, proactive rider safety platform. The system integrates multimodal sensing including physiological monitoring, EEG-based cognitive stress and fatigue

detection, IMU-driven risky event identification, and geospatial danger zone prediction within a cohesive edge-mobile architecture centered on an ESP32 processor and a Flutter-based companion application. By fusing data from biometric sensors (heart rate, temperature, single-channel EEG), motion sensing (IMU), and location tracking (GPS), the system delivers real-time assessment of cardiac risk, cognitive state, physical riding behavior, and route hazards. TensorFlow Lite models enable efficient on-device inference, while spatial indexing techniques support timely proximity alerts. The mobile application further enhances usability through live dashboards, pre-ride route risk analysis, automated post-ride reports, longitudinal analytics, and multilingual voice guidance tailored for the South Asian context. The proposed framework addresses key limitations in existing smart helmet solutions by providing unified, real-time monitoring and personalized analytics in a single accessible platform. Real-world prototype testing demonstrated the system's technical feasibility and strong potential for improving rider awareness and decision-making. Future work will focus on enhancing cognitive monitoring through multi-channel EEG, incorporating Vehicle-to-Everything (V2X) communication for broader situational awareness, exploring augmented reality helmet visor overlays, optimizing power consumption for extended operation, and pursuing regulatory certification for commercial deployment. Collaboration with road safety authorities, helmet manufacturers, and fleet operators will be essential to maximize real-world impact and contribute meaningfully to global efforts in reducing motorcycle-related fatalities.

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